

2

MODULE

DIAGNOSE

Where are you now?

Assessing your circular economy maturity, mapping resources,
and identifying hotspots for improvement

**CE Maturity
Assessment**

Baseline Mapping

Hotspot Analysis

1. Awareness

2. Diagnose

3. Options

4. Business Case

5. Implement

6. Monitor

What is the Diagnose Module?

Before acting, you need to understand your starting point. This module helps SMEs identify where they currently stand in their circular economy journey, assessing CE maturity, mapping resource flows, and locating key hotspots.

1 Maturity Assessment

Quick self-diagnosis of how circular your business already is across key dimensions.

Key outputs:

- CE maturity score
- Priority areas for focus

2 Baseline Mapping

Map your resource inputs, outputs, waste streams and energy flows systematically.

Key outputs:

- Input/output inventory
- Waste stream overview

3 Hotspot Analysis

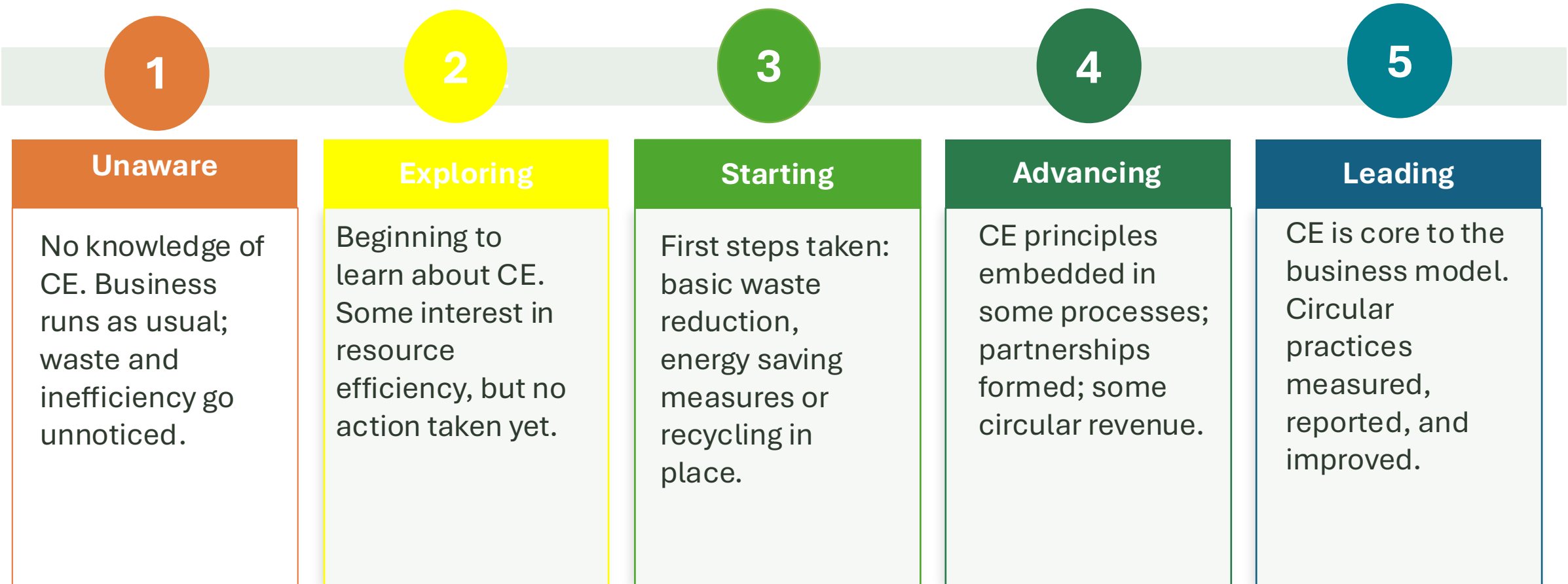
Pinpoint the most resource-intensive or wasteful activities in your operations.

Key outputs:

- Hotspot map
- Priorities for action

CE Maturity – Where Do You Think You Are?

Before using assessment tools, reflect on your current situation. Most SMEs fall somewhere along this maturity spectrum:



 *The tools in this module will help you confirm your position and identify priorities.*

How Can I Assess My Business?

Three complementary approaches, use one or all three depending on your needs and available time.



Tip: Start with the Quick Maturity Assessment if you are new to CE, then move on to Baseline Mapping and Hotspot Analysis for deeper insight.

CE Maturity Assessment Tools

Smart Circular Economy Assessment (SCEA)

Provider:	Ellen MacArthur Foundation
Time:	30–45 min
Preparation:	Basic business data
Purpose:	Helps SMEs benchmark their current CE level and understand key gaps.
Output:	CE maturity overview across 5 dimensions with visual output and priority recommendations.
Best for:	SMEs new to CE want a quick, accessible starting point.
Link:	ellenmacarthurfoundation.org/smart-material-choices

Learn to make smart material choices by asking the right questions. These steps will help you make better choices about what materials go into your products as well as their impact on the wider system.

I-GO Assessment Tool

Provider:	International Green Economy Association / GGKP
Time:	45–60 min
Preparation:	Project data, operations info
Purpose:	Assesses organisational readiness for resource efficiency and circularity.
Output:	Structured diagnosis of green economy readiness with scored indicators.
Best for:	SMEs with some sustainability activity looking for a more structured review.
Link:	https://igosolution.org/

I-GO helps SMEs become more productive, competitive, and resilient by making it easier to access tailored knowledge and support services on resource efficiency. It's the I-GO Assistant tool that guides businesses through self-assessment, suggests practical improvements, and connects them with customised support based on their specific needs.

Baseline Mapping

Understanding your resource flows

What is a Baseline?

A baseline captures your business as it is today: all materials entering and leaving, energy used, water consumed, and waste generated. It gives you a factual starting point to measure improvement over time.

Why Map Resources?

Many cost savings and circular opportunities are invisible until you map your flows. A simple resource map reveals which inputs cost the most, where waste is highest, and where efficiency gains are possible.

What to Map



Material inputs

Raw materials, components, packaging



Energy use

Electricity, fuel, heat by process



Water

Consumption and discharge



Waste streams

Type, quantity, disposal route



Product outputs

Volume, quality, by-products

💡 Tip: Start with a simple table listing all your inputs and outputs by category. Exact quantities are ideal but even rough estimates are a valuable starting point.

CE Maturity Assessment Tools

Smart Circular Economy Assessment (SCEA)

Provider:	Ellen MacArthur Foundation
Time:	30–45 min
Preparation:	Basic business data
Purpose:	Helps SMEs benchmark their current CE level and understand key gaps.
Output:	CE maturity overview across 5 dimensions with visual output and priority recommendations.
Best for:	SMEs new to CE want a quick, accessible starting point.
Link:	ellenmacarthurfoundation.org/smart-material-choices

Learn to make smart material choices by asking the right questions. These steps will help you make better choices about what materials go into your products as well as their impact on the wider system.

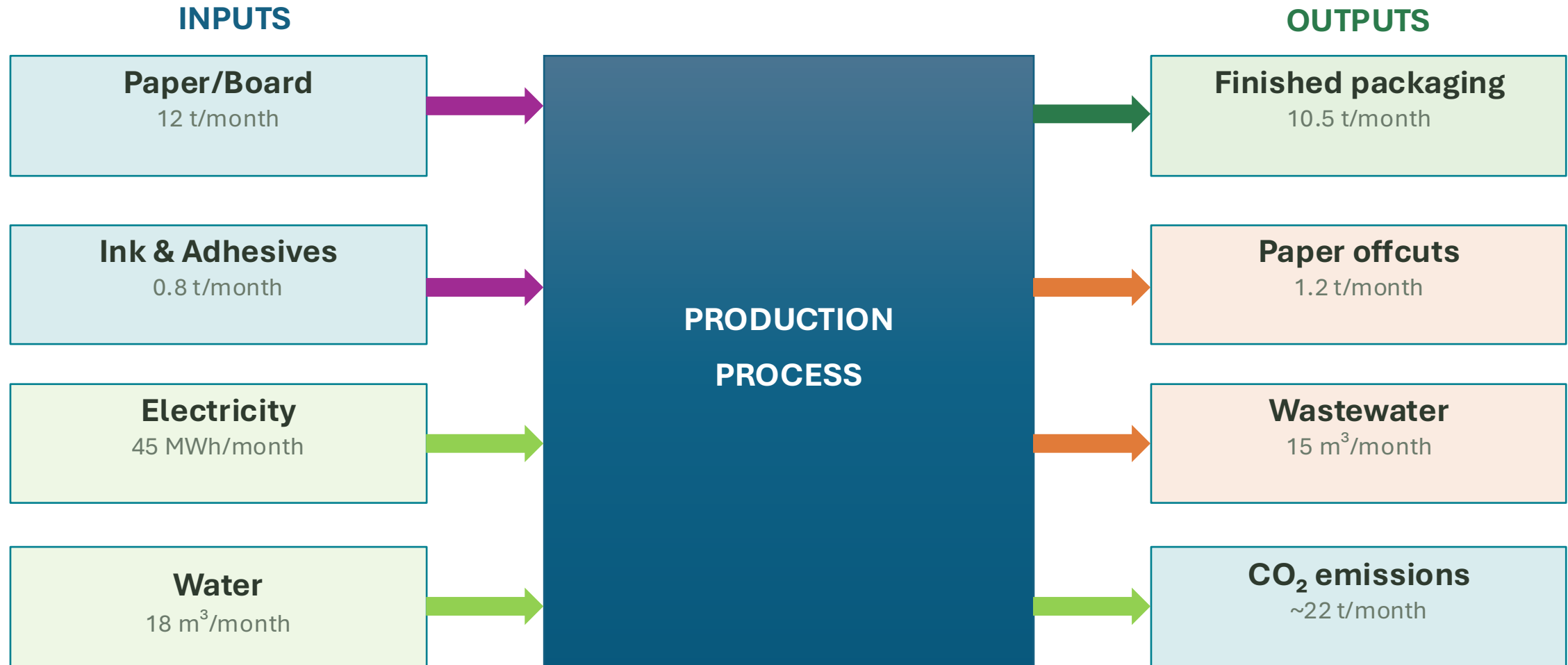
I-GO Assessment Tool

Provider:	International Green Economy Association / GGKP
Time:	45–60 min
Preparation:	Project data, operations info
Purpose:	Assesses organisational readiness for resource efficiency and circularity.
Output:	Structured diagnosis of green economy readiness with scored indicators.
Best for:	SMEs with some sustainability activity looking for a more structured review.
Link:	https://igosolution.org/

I-GO helps SMEs become more productive, competitive, and resilient by making it easier to access tailored knowledge and support services on resource efficiency. It's the I-GO Assistant tool that guides businesses through self-assessment, suggests practical improvements, and connects them with customised support based on their specific needs.

Example: Input/Output Map

GreenPack Ltd – small packaging manufacturer | Fictional illustrative example



Key insights from this map:

Paper offcuts (1.2 t/month) = lost raw material value | Wastewater could be partially treated and reused | Electricity is the highest-cost input – efficiency measures could save significantly

Baseline Mapping & Input/Output Tools

MODULE 2 DIAGNOSE

ready2LOOP Circularity Scan

Provider:	ready2LOOP
Time:	1–2 hours
Input quired:	Material, energy, and waste data
Purpose:	Maps material loops and identifies where circularity can be improved in production.
Output:	Circularity score, flow visualisation, improvement priorities
Best for:	Manufacturing SMEs with physical products
Link:	ready2LOOP Circularity Scan

The Circular Strategies Scanner helps manufacturing companies understand, map, and improve their circular economy strategies across products and operations. It is typically used in workshops to identify current initiatives, explore new opportunities for circularity, and support innovation and shared strategic vision.

Material Circularity Indicator (MCI)

Provider:	Ellen MacArthur Foundation
Time:	Half-day to 1 day
Input required:	Product design & material data, end-of-life data
Purpose:	Quantifies circularity at product or company level using a standardised metric.
Output:	MCI score (0–1) showing how circular a product or business is in practice
Best for:	Advanced users with detailed material data
Link:	Material Circularity Indicator

The Material Circularity Indicator (MCI) measures how effectively materials are kept in use throughout a product's life cycle and supports evidence-based circular economy decisions. The MCI calculator simplifies the original methodology, helping users turn circularity goals into clear, measurable results.

Hotspot Analysis

Identifying your most critical impacts and inefficiencies

What Is a Hotspot?

A hotspot is a point in your business where resource use, waste, cost, or environmental impact is disproportionately high. Hotspots are the priority areas where small changes can deliver the greatest circular economy benefits.



Energy Hotspots

Processes or equipment consuming disproportionate electricity, heat or fuel.



Material Hotspots

Materials that are expensive, scarce, or lost as waste in significant quantities.



Water Hotspots

Production stages consuming or polluting large volumes of water.

 Your hotspot analysis directly feeds into Module 3 (Options) – once you know where your biggest impacts are, you can identify the most impactful circular strategies to address them.

How to Identify Hotspots

- 1 Review your input/output map and highlight the largest flows
- 2 Flag activities with high energy or water use
- 3 Identify the costliest waste streams
- 4 Look for processes with high defect or rework rates
- 5 Consider supplier-side risks (price volatility, scarcity)
- 6 Gather staff insights (they often know where waste happens)

Hotspot Analysis Tools

Hotspot Analysis Tool (GGKP)

Provider: Green Growth Knowledge Platform

Time: Half day

Input: Resource use data, process list

Output: Ranked hotspot list with impact scores

Best for: General SMEs across sectors

Link: [Hotspot Analysis Tool \(GGKP\)](#)

The SCP-HAT Country Profile tool provides an overview of a country's environmental performance across key sustainable consumption and production policy areas, including materials use, climate change, energy, water, and pollution. It helps policymakers, NGOs, and the public identify sustainability hotspots and track environmental trends through indicators, data visualisations, and country-level analysis.

Saxion Quick-Scan (Circularity)

Provider: Saxion University of Applied Sciences

Time: 1–2 hours

Input: Business operations overview

Output: Circularity score + priority hotspots by business area

Best for: SMEs wanting a fast structured scan

Link: [Saxion Quick-Scan](#)

The Quicksan Circular Business Models (CBM) tool helps companies assess their current level of sustainable and circular entrepreneurship and explore opportunities for more circular business models. Through a questionnaire and interactive framework, it guides users in identifying suitable circular strategies, organisational structures, and revenue models to support circular business innovation.

Future Adaptive Design Guide

Provider: RISE Research Institutes of Sweden

Time: Flexible – workshop format

Input: Product or process design information

Output: Design-focused hotspots; adaptation opportunities for circularity

Best for: Product design and manufacturing SMEs

Link: [ri.se \(2023 publication\)](#)

Future adaptive design helps companies reduce business risks in circular business models by creating durable, flexible, and upgradeable products that remain useful and attractive over longer lifetimes. The approach provides tools and methods to support circular business innovation, improve profitability, and identify strategies for extending product life while reducing environmental impact.

These tools can be used in sequence: use the GGKP or Saxion tool to identify hotspots, then the Future Adaptive Design Guide to explore circular design solutions.

From Mapping to Action: A Construction SME

Company Profile

Sector:	Construction & renovation
Size:	12 employees, Ukraine
Activity:	Residential renovation, flooring & insulation
CE Experience:	None before this assessment

Step 1

Maturity Assessment

Scored 1.8/5 – mostly unaware of CE; no waste tracking, no recycling partnerships.

Step 2

Baseline Mapping

Discovered 2.1 tonnes of waste board and timber off-cuts per month – worth ~€800.

Step 3

Hotspot Identified

Insulation off-cuts were the largest waste stream and could be sold to a local reseller.

Additional Resources & Further Reading

Deepen your knowledge of CE assessment with these freely available resources:

Foundation Knowledge

Ellen MacArthur Foundation – Core CE concepts

Comprehensive introduction to CE principles, the butterfly model, and case studies.

 ellenmacarthurfoundation.org

Metrics & Measurement

EMF – Metrics for Circular Business Models

Guidance on selecting indicators to make the case for circular business models.

 ellenmacarthurfoundation.org/metrics

Tools & Databases

Green Growth Industry Platform (GGKP)

Searchable database of tools, policies, and resources for the green economy.

 search.ggkp.org

UNEP – Resource Efficiency Toolkit

Practical guides for SMEs on resource efficiency, waste reduction and circular strategies.

 unep.org/resources

WBCSD – Circular Transition Indicators

A framework for measuring corporate circularity with SME-appropriate indicators.

 wbcsd.org/cti

Nordic Council CE Case Studies

Case studies of SMEs implementing CE in the Nordic/Baltic region – strong examples.

 norden.diva-portal.org

Tip: Save these links and share them with your team before starting the assessment exercises.

2

Diagnose – Summary

MODULE

What you have covered in this module:

- ✓ Understood what the Circular Economy is and why it matters for your business
- ✓ Identified your current CE maturity level (1–5 scale)
- ✓ Mapped your resource inputs, outputs, and waste streams
- ✓ Identified your main hotspots: energy, materials, water
- ✓ Explored practical tools: SCEA, I-GO, ready2LOOP, MCI, Saxion Scan
- ✓ Reviewed a real-world case example and potential savings

Next Steps

- 1 Complete assessment**
Use the SCEA or I-GO tool to confirm your CE maturity score.
- 2 Build your resource map**
Fill in the input/output template provided in the toolkit.
- 3 Mark your hotspots**
Highlight 2–3 priority areas from your resource map.
- 4 Move to Module 3**
Take your hotspots into the Options module to identify CE strategies.

1 Awareness

2 Diagnose

3 Options

4 Business Case

5 Implement

6 Monitor